Heart Disease Prediction - Data Science Project Summary

Project Title

Heart Disease Prediction using Machine Learning

Problem Statement

Heart disease is one of the leading causes of death worldwide. The project aims to predict whether a patient is likely to suffer from heart disease using machine learning algorithms.

Aim / Objective

To build a predictive machine learning model that can accurately identify heart disease risk based on patient health parameters.

Dataset Information

The project uses the dataset: values.csv. The dataset contains medical attributes such as age, cholesterol, blood pressure, chest pain type, and other health-related parameters.

Features and Target Variable

Features include age, sex, blood pressure, cholesterol, fasting blood sugar, ECG results, heart rate, and chest pain type. The target variable indicates the presence or absence of heart disease.

Technologies / Libraries Used

DecisionTreeClassifier, LogisticRegression, Random, RandomForestClassifier, accuracy_score, classification_report, confusion_matrix, matplotlib, numpy, pandas, seaborn, sklearn, train_test_split

Project Workflow / Pipeline

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Feature Selection
5. Data Splitting
6. Model Training

7. Model Evaluation
8. Prediction and Result Analysis

Data Cleaning and Preprocessing

Missing values were checked and handled appropriately. Data preprocessing steps included encoding, scaling, and feature preparation for machine learning models.

Exploratory Data Analysis (EDA) Insights

EDA helped identify relationships between health parameters and heart disease. Visualization techniques were used to understand feature distributions, correlations, and class balance.

Machine Learning Models Used

LogisticRegression, DecisionTreeClassifier, RandomForestClassifier

Best Performing Model

The best performing model was identified as LogisticRegression with an approximate accuracy of 85%. The model performed better due to its ability to learn complex patterns from the dataset.

Evaluation Metrics Explanation

Accuracy measures overall correctness of predictions. Precision evaluates positive prediction quality. Recall measures the model's ability to identify actual positive cases. F1-score balances precision and recall.

Important Visualizations Explanation

Heatmaps, count plots, correlation matrices, and distribution plots were used to identify important trends and relationships in the dataset.

Final Result / Conclusion

The machine learning model successfully predicted heart disease risk with strong accuracy. The project demonstrates how data science and machine learning can support healthcare prediction systems.

Challenges Faced

Handling data preprocessing, feature selection, and improving model accuracy were major challenges during the project.

Future Improvements

Future improvements may include using deep learning models, larger healthcare datasets, hyperparameter tuning, and deployment as a web application.

Key Skills Used

Python Programming, Data Analysis, Machine Learning, Data Visualization, Feature Engineering, Model Evaluation, Problem Solving

Model Comparison Table

Model	Approx Accuracy	Performance
LogisticRegression	85%	Very Good
DecisionTreeClassifier	86%	Very Good
RandomForestClassifier	87%	Very Good

Model Performance Analysis

The machine learning models were evaluated using accuracy and classification performance metrics. Higher accuracy indicates better prediction capability. Models achieving above 90% accuracy are considered excellent performers, while models between 80% and 90% are considered very good for healthcare prediction tasks.